NARULA INSTITUTE OF TECHNOLOGY
An Autonomous Institute under MAKAUT

B.TECH./CSE, IT, EE, ECE, EIE, CE, ME, CST, CSE(AI-ML)/ODD/Sem-1/R21/M101/2022-2023
YEAR: 2023

MATHEMATICS-I
M101

TIME ALLOTTED: 3 HOURS FULL MARKS: 70

The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable

GROUP – A
(Multiple Choice Type Questions)

1. Answer any ***ten*** from the following, choosing the correct alternative of each question: **10×1=10**

(i)	The Eigen values of the matrix is $\begin{pmatrix} 1 & 4 \\ 4 & 1 \end{pmatrix}$ are a) $-5, -3$ b) $-5, 3$ c) $5, -3$ d) $5, 3$	1	2	Understanding
(ii)	$f(x, y) = \sin^{-1} \frac{x^3 + y^3}{x + y}$ is a homogeneous function of degree a) 1 b) 0 c) 2 d) $1/2$	1	2	Understanding
(iii)	If Cauchy's mean value theorem is applied on two functions $f(x) = x$ and $g(x) = x^2$ in the interval $[-1, 1]$, then c equals to a) 1 b) 0 c) -1 d) $\frac{1}{2}$	1	2	Understanding
(iv)	If for a matrix $A_{3\times3}$, $Rank(A) = 2$, then the matrix A is a) singular b) non-singular c) orthogonal d) skew-symmetric	1	2	Understanding
(v)	Which of the following functions does not satisfy Rolle's theorem in$[-2,2]$ interval a) $\frac{1}{x-3}$	1	2	Understanding

b) $\frac{1}{x}$
c) x
d) x^2

(vi) $\int_0^2 \int_y^{y^2} xy\,dx\,dy =$ 1 2 Understanding

a) $-\frac{1}{15}$
b) $\frac{28}{15}$
c) $\frac{13}{8}$
d) $\frac{19}{17}$

(vii) The sum of the series $\frac{1}{1.2}+\frac{1}{2.3}+\frac{1}{3.4}+\cdots$is 1 2 Understanding

a) 1
b) 0
c) $1/2$
d) does not exist

(viii) $\lim_{n\to\infty}\frac{4n^2+6n-1}{2n^2-1} =$ 1 2 Understanding

a) 1
b) 2
c) 3
d) 4

(ix) The value of m such that $\vec{A}=(mxy-z^3)\hat{\imath}+(m-2)x^2\hat{\jmath}+(1-m)xz^2\hat{k}$ is irrotational is 1 2 Understanding

a) 0
b) 4
c) -4
d) 2

(x) The directional derivative of $f=xy+yz+zx$ in the direction vector $\hat{\imath}+2\hat{\jmath}+2\hat{k}$ at the point (1,2,0) 1 2 Understanding

a) $\frac{10}{3}$
b) $-\frac{1}{6}$
c) $-\frac{10}{3}$
d) $-\frac{1}{3}$

(xi) If $A=\begin{bmatrix}1 & -2 & 3\\ 0 & 1 & 4\\ 2 & -1 & 0\end{bmatrix}$then the product of the Eigen values of A is 1 1 Remember

a) -12

b) 18
c) -18
d) 24

(xii) Two Eigen values of the matrix $A = \begin{bmatrix} 2 & -2 & 2 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{bmatrix}$ are 1 and -1. The third eigen value of A is | 1 | 1 | Remember

a) 1
b) 0
c) 3
d) 2

GROUP – B
(Short Answer Type Questions)
(Answer any three of the following) 3 x 5 = 15

2.	Find the rank of the matrix, $A = \begin{pmatrix} 1 & 3 & 2 & 4 & 1 \\ 0 & 0 & 2 & 2 & 0 \\ 2 & 6 & 2 & 6 & 2 \\ 3 & 9 & 1 & 10 & 6 \end{pmatrix}$	5	3	Applying
3.	Use Mean value theorem prove that $0 < \frac{1}{\log(1+x)} - \frac{1}{x} < 1$, for x> 0.	5	3	Applying
4.	If $u = \sin^{-1}\left(\frac{x^2+y^2}{x+y}\right)$, show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \tan u$.	5	3	Applying
5.	Evaluate $\iiint_C (x+y+z+1)^4 dxdydz$, where C is the region defined by $x \geq 0, y \geq 0, z \geq 0, x+y+z \leq 1$.	5	5	Evaluating
6.	Find gradient of the function $f(x,y,z) = x^2yz + 4xz^2$ at the point $(1,-2,-1)$.	5	5	Evaluating

GROUP – C
(Long Answer Type Questions)
(Answer any three of the following) 3 x 15 = 45

7. **(i)** Determine the condition involving a and b for which the following system of equations | 8 | 5 | Evaluating

$$\begin{aligned} x + y + z &= b \\ 2x + y + 3z &= b + 1 \\ 5x + 2y + az &= b^2 \end{aligned}$$

		admits infinite solutions.			
	(ii)	If $$f(x,y)=\begin{cases}\frac{xy(x^2-y^2)}{x^2+y^2} & ,\ (x,y)\neq(0,0)\\ 0 & ,\ (x,y)=(0,0)\end{cases}$$ Verify whether $f_{xy}(0,0)=f_{yx}(0,0)$.	7	2	Understanding
8.	**(i)**	Diagonalize the following matrix (if possible) $$A=\begin{bmatrix}3 & 1 & -1\\ 2 & 2 & -1\\ 2 & 2 & 0\end{bmatrix}$$	5	2	Understanding
	(ii)	For $u(x,y)=tan^{-1}\frac{x^2+y^2}{x+y}$, show that $x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=\frac{1}{2}sin2u$	5	3	Applying
	(iii)	Test the convergence of the following series $$\frac{1^2}{2^2}+\frac{1^2.3^2}{2^2.4^2}+\frac{1^2.3^2.5^2}{2^2.4^2.6^2}+\cdots$$	5	4	Analyzing
9.	**(i)**	If $u=f(\frac{y-x}{xy},\frac{z-x}{zx})$, show that $x^2\frac{\partial u}{\partial x}+y^2\frac{\partial u}{\partial y}+z^2\frac{\partial u}{\partial z}=0$	5	3	Applying
	(ii)	Evaluate: $\int_0^{log2}\int_0^{x}\int_0^{x+y}e^{x+y+z}dxdydz$	5	5	Evaluating
	(iii)	If $\vec{a}=x^2y\hat{\imath}+y^2z\hat{\jmath}+z^2x\hat{k}$, find $\vec{\nabla}.\vec{a}$ at the point $(0,1,1)$.	5	3	Applying
10.	**(i)**	If $f(x)=\begin{cases}x^{\alpha}logx: & x>0\\ 0: & x=0\end{cases}$, find the value of α, so that $f(x)$ obeys Rolle's theorem in $[0,1]$.	8	4	Analyzing
	(ii)	Test the convergence of the series $\frac{1}{1.2^2}+\frac{1}{2.3^2}+\frac{1}{3.4^2}+\ldots$	7	4	Analyzing
11.	**(i)**	Find the maximum and minimum values of the function $x^3+y^3-3x-12y-20$.	9	3	Applying
	(ii)	For $u(x,y)=\log(tanx+tany)$, show that $\frac{\partial u}{\partial x}sin2x+\frac{\partial u}{\partial y}sin2y=2$	6	3	Applying